

*R*DataMining

Class 04 R Lab

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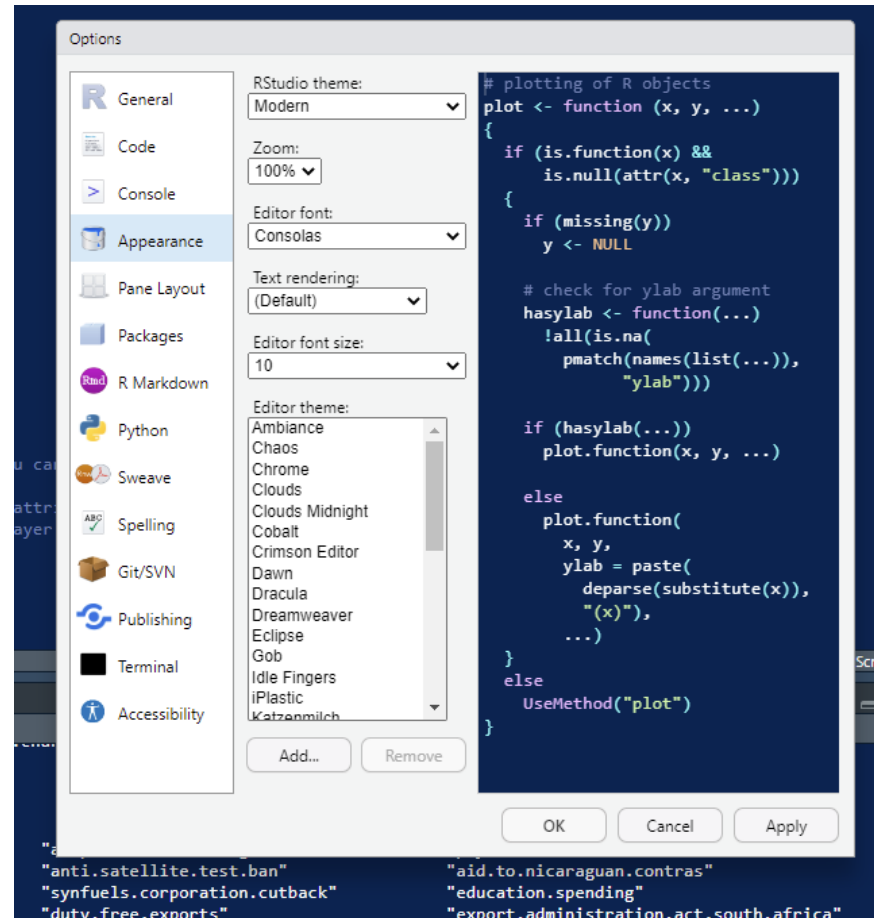
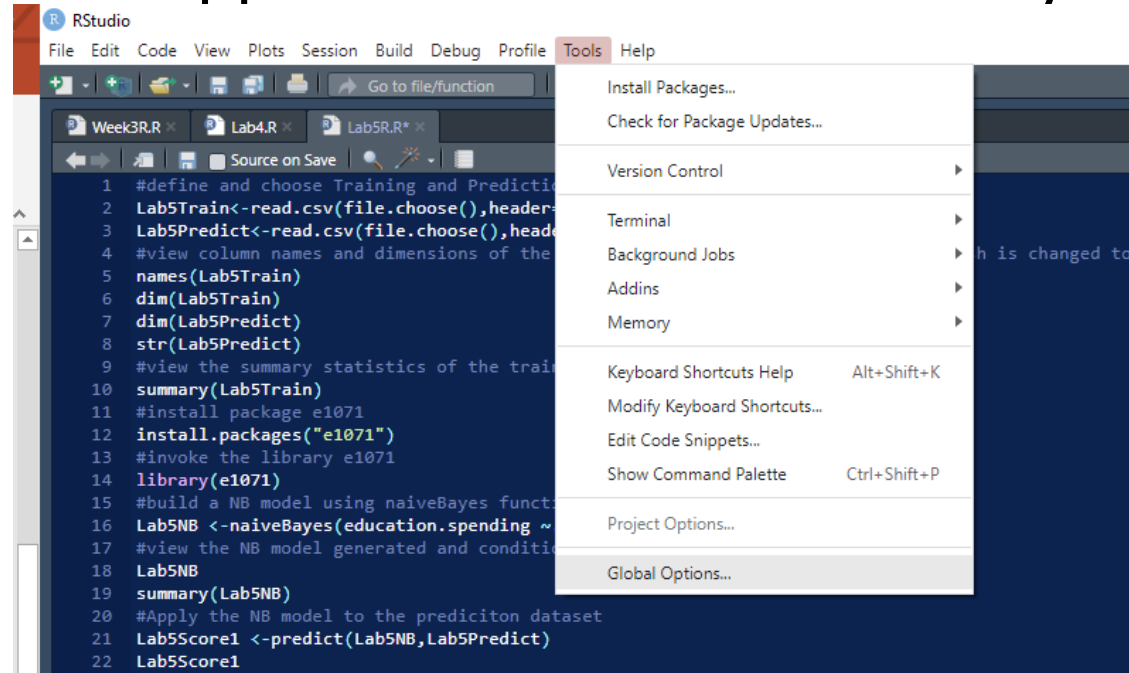


Instructions

- R scripts for Naïve Bayes classifier and Logistic Regression classifier are provided in the next few slides, please follow them and complete the lab in R.
- You are recommended to type notes (starting at #) as it's a good manner to have them in you code.
- In order to see codes and notes clearly, I show the script in RStudio.
- There are five questions in this lab to answer. Please type your answer via HW2 Submission.

Change appearance in RStudio

If you prefer a difference appearance, you can change it: Tools → Global Options → Appearance. You can choose your favorite text size, editor theme, etc.



Naïve Bayes Classifier

```
1 #define and choose Training and Prediction datasets
2 Lab5Train<-read.csv(file.choose(),header=T,stringsAsFactors = T)
3 Lab5Predict<-read.csv(file.choose(),header=T,stringsAsFactors = T)
4 #view column names and dimensions of the training dataset (please notice that dash is changed to dot in R)
5 names(Lab5Train)
6 dim(Lab5Train)
7 #view the summary statistics of the training dataset
8 summary(Lab5Train)
9 #install package e1071
10 install.packages("e1071")
11 #invoke the library e1071
12 library(e1071)
13 #build a NB model using naiveBayes function and including all attributes
14 Lab5NB <-naiveBayes(education.spending ~., data=Lab5Train)
15 #view the NB model generated and conditional probabilities
16 Lab5NB
17 #Apply the NB model to the prediction dataset
18 Lab5Score <-predict(Lab5NB,Lab5Predict)
19 #view the prediction result
20 Lab5Score
21 #show the summary statistics of Lab5Score
22 summary(Lab5Score)
23 #the following three rows of script help us write the prediction value of education.spending to the prediction dataset
24 Score<-data.frame(Lab5Score)
25 Lab5Predict$education.spending<-Score$Lab5Score
26 Lab5Predict
```

Here, we ask R to convert all the strings to factors by adding a new parameter, `stringsAsFactors = True`; otherwise, they will be characters and then you have to convert them to factors later.

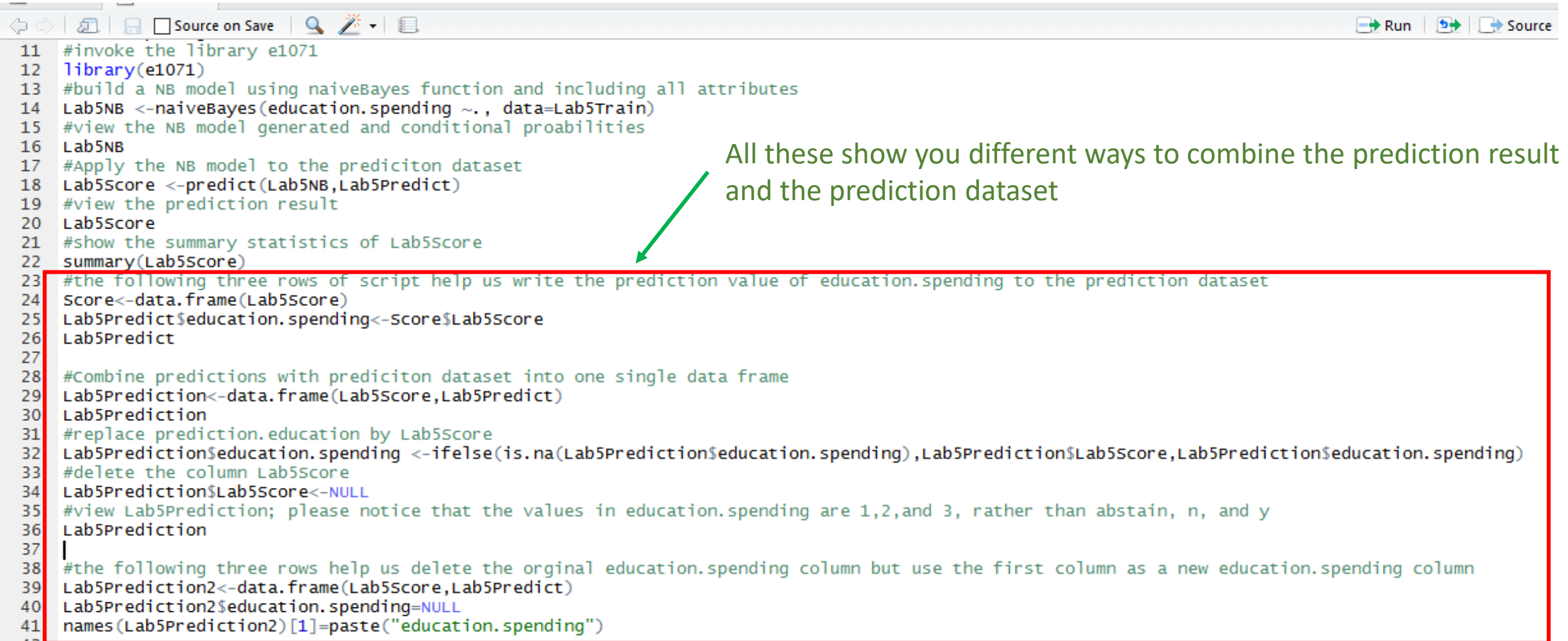
You can also install the package “rminer”(see details [here](#)) as an alternative

For details about naïve classifier, please click [here](#)

Q1: Please indicate how many of them are predicted to vote as y.

Q2: We practice Naïve Bayes classifier in both R and RapidMiner, do they generate the same prediction for each record in the prediction dataset? (Hint: you can list predictions results generated by the two methods in Excel and then check if the prediction results are different in each row; export the prediction result using `write.csv()` function.

Additional Script (Not Required)



```
11 #invoke the library e1071
12 library(e1071)
13 #build a NB model using naiveBayes function and including all attributes
14 Lab5NB <-naiveBayes(education.spending ~., data=Lab5Train)
15 #view the NB model generated and conditional probabilities
16 Lab5NB
17 #Apply the NB model to the prediction dataset
18 Lab5Score <-predict(Lab5NB,Lab5Predict)
19 #view the prediction result
20 Lab5Score
21 #show the summary statistics of Lab5Score
22 summary(Lab5Score)
23 #the following three rows of script help us write the prediction value of education.spending to the prediction dataset
24 Score<-data.frame(Lab5Score)
25 Lab5Predict$education.spending<-Score$Lab5Score
26 Lab5Predict
27
28 #Combine predictions with prediction dataset into one single data frame
29 Lab5Prediction<-data.frame(Lab5Score,Lab5Predict)
30 Lab5Prediction
31 #replace prediction.education by Lab5Score
32 Lab5Prediction$education.spending <-ifelse(is.na(Lab5Prediction$education.spending),Lab5Prediction$Lab5Score,Lab5Prediction$education.spending)
33 #delete the column Lab5Score
34 Lab5Prediction$Lab5Score<-NULL
35 #view Lab5Prediction; please notice that the values in education.spending are 1,2,and 3, rather than abstain, n, and y
36 Lab5Prediction
37 |
38 #the following three rows help us delete the original education.spending column but use the first column as a new education.spending column
39 Lab5Prediction2<-data.frame(Lab5Score,Lab5Predict)
40 Lab5Prediction2$education.spending=NULL
41 names(Lab5Prediction2)[1]=paste("education.spending")
```

All these show you different ways to combine the prediction result and the prediction dataset

Show Conditional a-posterior probabilities in R (not required)

```
5 names(Lab5Train)
6 dim(Lab5Train)
7 #view the summary statistics of the training dataset
8 summary(Lab5Train)
9 #install package e1071
10 install.packages("e1071")
11 #invoke the library e1071
12 library(e1071)
13 #build a NB model using naiveBayes function and including all attributes
14 Lab5NB <-naiveBayes(education.spending ~., data=Lab5Train)
15 #view the NB model generated and conditional probabilities
16 Lab5NB
17 #Apply the NB model to the prediction dataset
18 Lab5Score <-predict(Lab5NB,Lab5Predict)
19
20 #the following two rows of scripts are used to generate the conditional a-posterior probabilities for each class, and the class with maximal probability else.
21 Lab5Score2 <-predict(Lab5NB,Lab5Predict,type = "raw")
22 Lab5Score2|
23
24 #view the prediction result
25 Lab5Score
26 #show the summary statistics of Lab5Score
27 summary(Lab5Score)
28 #the following three rows of script help us write the prediction value of education.spending to the prediction dataset
29 Score<-data.frame(Lab5Score)
30 Lab5Predict$education.spending<-Score$Lab5Score
31 Lab5Predict
32
```

When you add type="raw" as an additional argument, you can see the conditional a-posterior probabilities for each class, and the class with maximal probability else.

Logistic Regression Classifier

```
# change the value "abstain" to "n" to make the variable binominal
Lab5Train$education.spending[Lab5Train$education.spending=="abstain"]<-"n"
# develop a logistic regression model using glm function.
LogModel <- glm(education.spending ~ .,family = "binomial", data=Lab5Train)
#view the logistic regression model
summary(LogModel)
#use the LR model to make prediction
Lab5ScoreLR <-predict(LogModel,Lab5Predict,type="response")
#view the prediciton; Round all those predicted probabilities to the third decimal place.
round(Lab5ScoreLR, 3)
#check if each predicted probablity is greater than 0.5 (i.e., with y as the predicted class)
Lab5ScoreLR>0.5
# count how many of them are predicted as y (i.e., probability greater than 0.5)
sum(Lab5ScoreLR>0.5)
```

For more details about the glm function, please check this [link](#).

Q3: Please indicate how many of them are predicted to vote as y.

The reason why we chose type="response" is that the default predictions of the binomial Logistic Regression are log-odds, and type = "response" gives the predicted probabilities. For details, please check [this link](#).

Q4: After you run round(Lab5ScoreLR, 3) in R, please compare each record's probability of voting y in R with that in RM (i.e., Confidence(y) in RM prediction result). Do the two tools generate the same probability of voting y?

Q5: Do R and RM generate the same prediction (y or n) for each record in the prediction dataset?

FAQs

- After you import your training and prediction dataset, please check their structure. Because we apply the **stringAsFactors = True**, all the columns except the target attribute in the prediction dataset should be factors. If you had characters, you may have to convert them to factors; otherwise, it might make your prediction results differently.

```
> str(Lab5Predict)
'data.frame': 35 obs. of 17 variables:
 $ handicapped.infants : Factor w/ 3 levels "abstain","n",...: 2 2 1 2 3 2 3 2 3 2 ...
 $ water.project.cost.sharing : Factor w/ 3 levels "abstain","n",...: 3 3 2 3 3 2 2 2 2 2 ...
 $ adoption.of.the.budget.resolution : Factor w/ 3 levels "abstain","n",...: 2 2 3 2 2 2 3 2 3 2 ...
 $ physician.fee.freeze : Factor w/ 2 levels "n","y": 2 2 2 2 2 2 1 2 1 2 ...
 $ el.salvador.aid : Factor w/ 3 levels "abstain","n",...: 3 3 2 3 3 3 3 3 2 3 ...
 $ religious.groups.in.schools : Factor w/ 3 levels "abstain","n",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ anti.satellite.test.ban : Factor w/ 3 levels "abstain","n",...: 2 2 3 2 2 2 2 2 3 2 ...
 $ aid.to.nicaraguan.contras : Factor w/ 3 levels "abstain","n",...: 1 2 3 2 2 2 2 2 3 2 ...
 $ mx.missile : Factor w/ 3 levels "abstain","n",...: 2 2 3 2 2 2 3 2 3 2 ...
 $ immigration : Factor w/ 3 levels "abstain","n",...: 2 3 3 3 3 2 3 2 2 2 ...
 $ synfuels.corporation.cutback : Factor w/ 3 levels "abstain","n",...: 1 3 2 2 2 2 2 3 2 2 ...
 $ education.spending : logi NA NA NA NA NA NA ...
 $ superfund.right.to.sue : Factor w/ 3 levels "abstain","n",...: 1 3 2 1 3 3 3 3 1 3 ...
 $ crime : Factor w/ 3 levels "abstain","n",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ duty.free.exports : Factor w/ 3 levels "abstain","n",...: 2 2 2 2 2 2 2 2 3 2 ...
 $ export.administration.act.south.africa: Factor w/ 3 levels "abstain","n",...: 1 2 3 2 3 3 3 2 3 2 ...
 $ Party : Factor w/ 2 levels "democrat","republican": 2 2 2 2 2 2 1 1 1 2 ...
```

```
> str(Lab5Train)
'data.frame': 400 obs. of 17 variables:
 $ handicapped.infants : Factor w/ 3 levels "abstain","n",...: 2 2 1 2 3 2 2 2 2 3 ...
 $ water.project.cost.sharing : Factor w/ 3 levels "abstain","n",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ adoption.of.the.budget.resolution : Factor w/ 3 levels "abstain","n",...: 2 2 3 3 3 3 2 2 2 3 ...
 $ physician.fee.freeze : Factor w/ 3 levels "abstain","n",...: 3 3 1 2 2 2 3 3 3 2 ...
 $ el.salvador.aid : Factor w/ 3 levels "abstain","n",...: 3 3 3 1 3 3 3 3 3 2 ...
 $ religious.groups.in.schools : Factor w/ 3 levels "abstain","n",...: 3 3 3 3 3 3 3 3 3 2 ...
 $ anti.satellite.test.ban : Factor w/ 3 levels "abstain","n",...: 2 2 2 2 2 2 2 2 2 3 ...
 $ aid.to.nicaraguan.contras : Factor w/ 3 levels "abstain","n",...: 2 2 2 2 2 2 2 2 2 3 ...
 $ mx.missile : Factor w/ 3 levels "abstain","n",...: 2 2 2 2 2 2 2 2 2 3 ...
 $ immigration : Factor w/ 3 levels "abstain","n",...: 3 2 2 2 2 2 2 2 2 2 ...
 $ synfuels.corporation.cutback : Factor w/ 3 levels "abstain","n",...: 1 2 3 3 3 2 2 2 2 2 ...
 $ education.spending : Factor w/ 3 levels "abstain","n",...: 3 3 2 2 1 2 2 2 3 2 ...
 $ superfund.right.to.sue : Factor w/ 3 levels "abstain","n",...: 3 3 3 3 3 3 1 3 3 2 ...
 $ crime : Factor w/ 3 levels "abstain","n",...: 3 3 3 2 3 3 3 3 3 2 ...
 $ duty.free.exports : Factor w/ 3 levels "abstain","n",...: 2 2 2 2 3 3 3 1 2 1 ...
 $ export.administration.act.south.africa: Factor w/ 3 levels "abstain","n",...: 3 1 2 3 3 3 3 3 3 1 ...
 $ Party : Factor w/ 2 levels "democrat","republican": 2 2 1 1 1 1 1 2 2 1
```


- You can use `as.factor()` to convert to a factor column, but in our case, we have multiple variables to convert. It will be much easier to use the following codes.
- **# convert character columns in the training set and prediction set to factor columns.**

```
Lab5Train[sapply(Lab5Train, is.character)] <- lapply(Lab5Train[sapply(Lab5Train,  
is.character)], as.factor)
```

```
Lab5Predict[sapply(Lab5Predict, is.character)] <- lapply(Lab5Predict[sapply(Lab5Predict,  
is.character)], as.factor)
```

- Then, you can try `str()` again to see if both of them are converted.
- Further Readings:
 - <https://stackoverflow.com/questions/20637360/convert-all-data-frame-character-columns-to-factors>
 - <https://statisticsglobe.com/convert-character-to-factor-in-r>
 - <https://stackoverflow.com/questions/30248583/error-could-not-find-function>